MC Fundamentals Task Sheet

Attachment Measure Up

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Learning Objective

Users will be able to apply select measurements to add an attachment to the machine and perform attachment movements to ensure the EW system can take readings to plumb the bottom of attachment and determine G-J measurement.

The precise measure-up of a hydraulic excavator attachment is part of the configuration of a Trimble Earthworks excavator system. The measure-up procedure outlined in the Trimble Earthworks Excavator Commissioning Manual is the most widely practiced. In the exercise below, you will use the Web interface Install Assistant on a real world machine to perform the attachment measure-up procedure that is used when adding a new attachment in the Web interface.

You will also learn how to adjust for cutting edge wear and edit bucket dimensions from the Earthworks Application.

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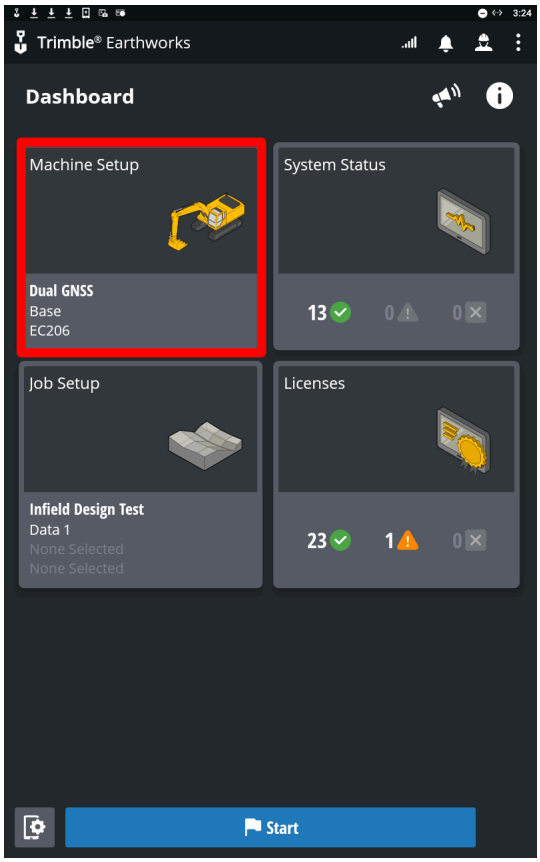
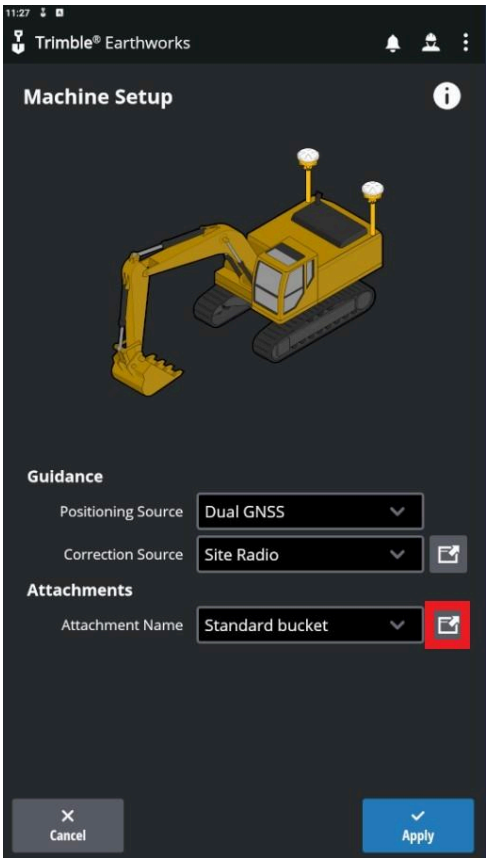
Attachment Measure-Up

Adding an attachment via the Trimble Earthworks Operator Application is an option available to operators depending on the permissions they have been granted. This task sheet will focus on accessing the attachments menu via the Machine Setup Tile.

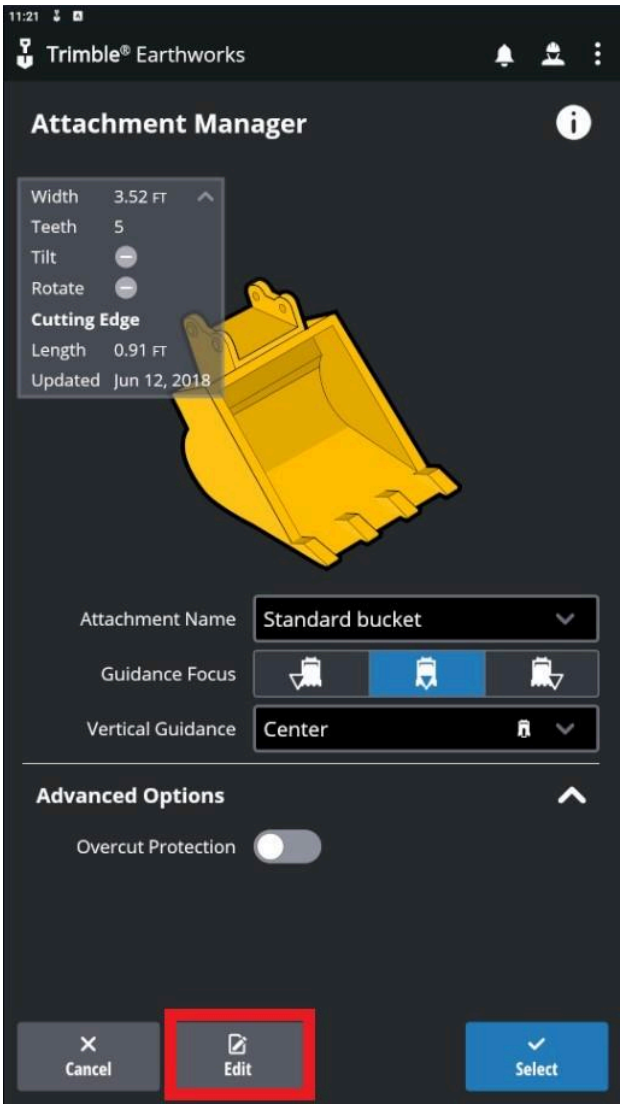
Note- Operator must be granted Operator Plus access to be able to complete this task

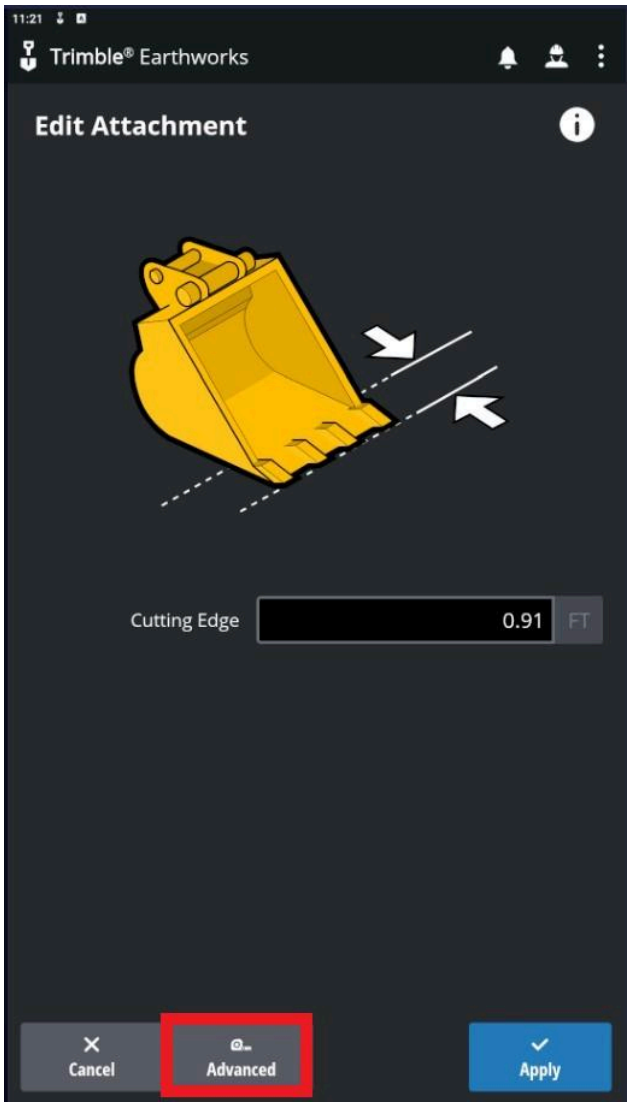
1. Open the Machine Settings tile and enter the Attachment Manager screen. Select the icon next to Attachment to enter Attachment Manager.



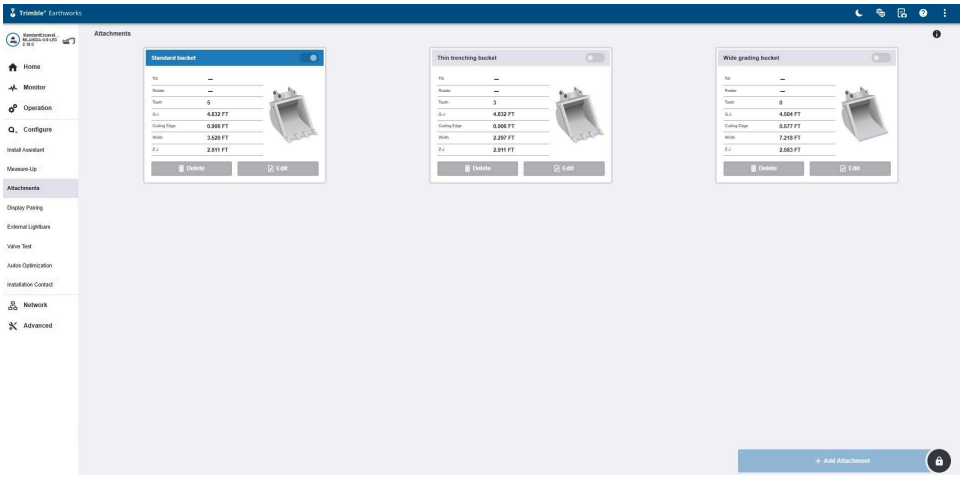


2. Within Attachment Manager, Select the Edit button to edit the attachment. Select the Advanced button to configure a new attachment.





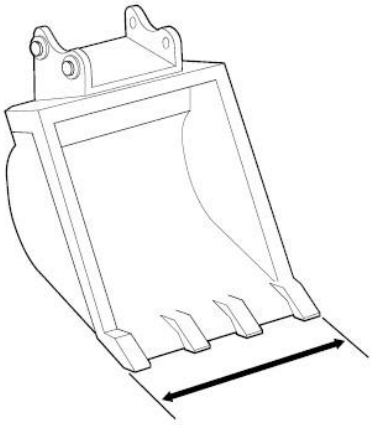
3. Earthworks will then open the Web Interface. This feature will only allow the operator access to the Attachments section within the WebUI.



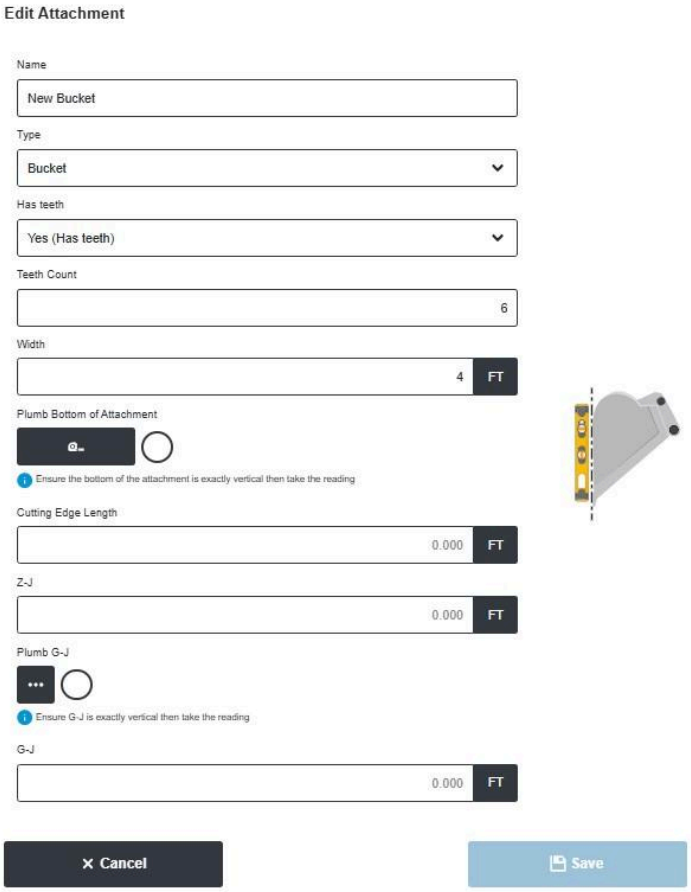
4. Select + Add Attachment to begin the attachment measure up process
5. Name the attachment with your name and date. Select Type of Attachment - Bucket and number of teeth.

Note: The procedure for adding a tilt bucket or Tilt Rotator can be found in the Commissioning Manual

6. Measure and enter the width of the cutting edge. This is generally measured to the widest part of the attachment.



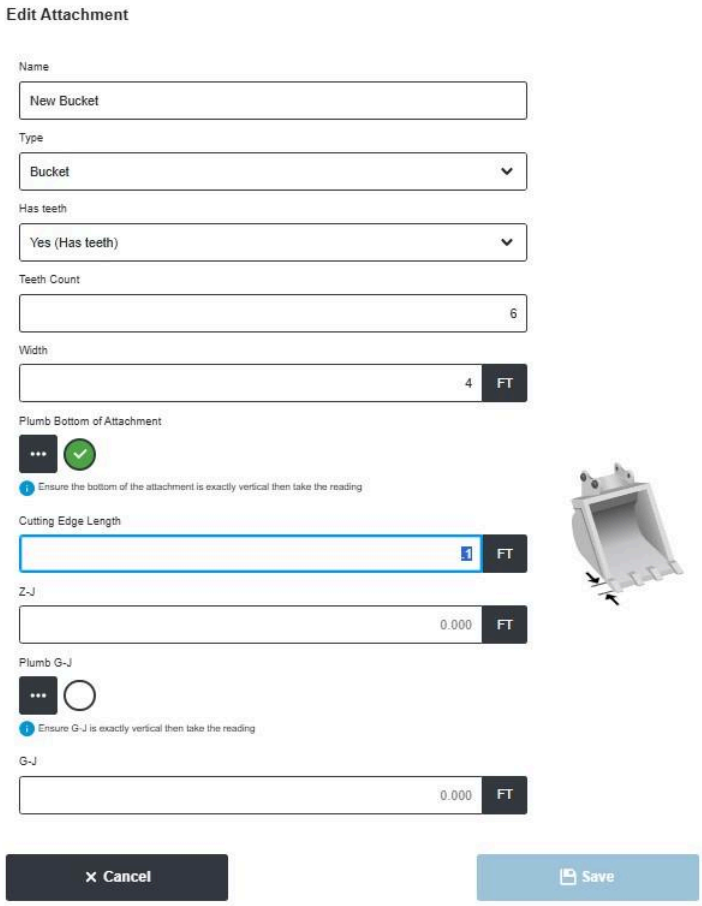
7. Plumb bottom of attachment - move the bucket so the bottom is vertical as pictured in the WebUI graphic.

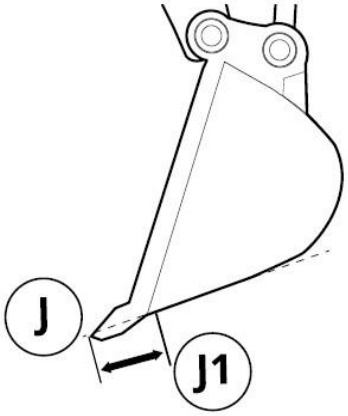


Note: You may need to extend the stick or uncurl/curl the bucket to ensure the bottom is completely vertical

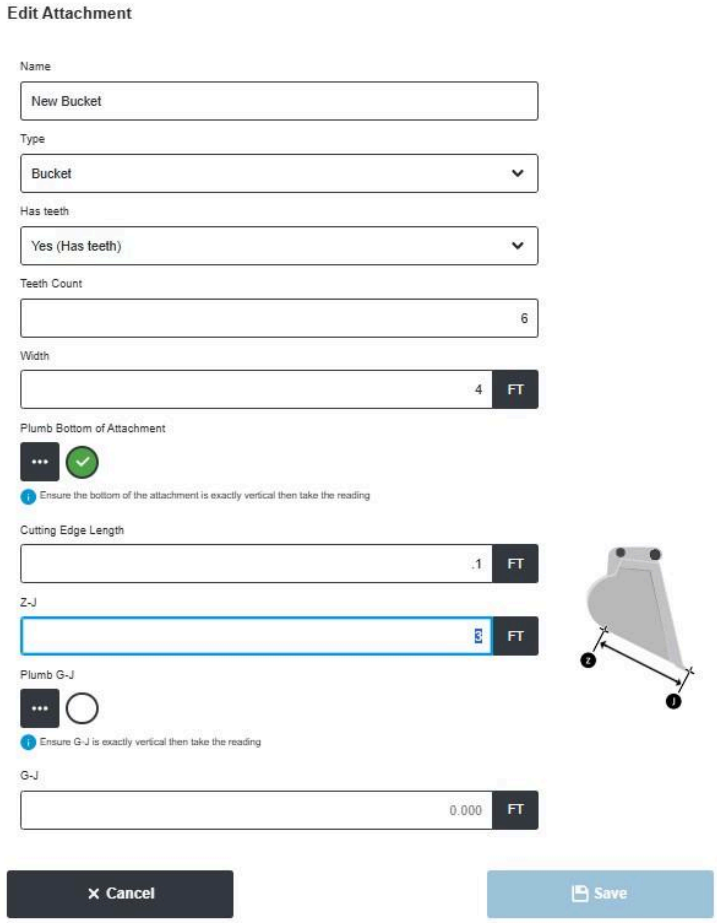
8. Tap the tape measure button and wait for the system to confirm value with a green check mark.

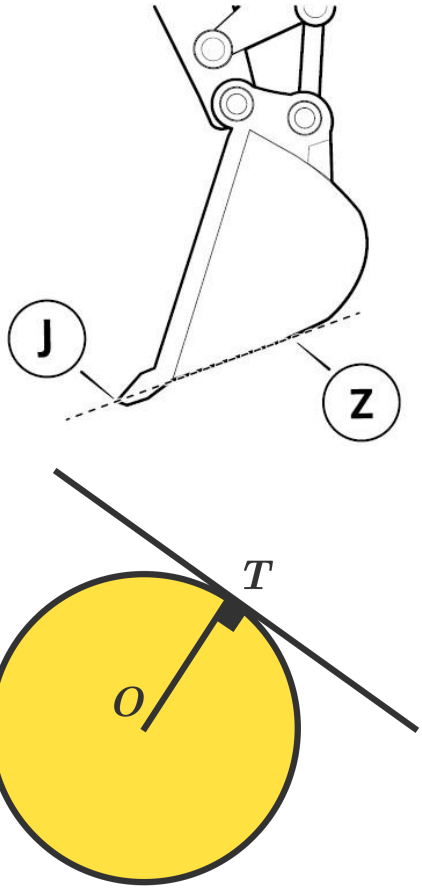
9. Measure the Cutting Edge Length and enter the dimension. The measurement is made from J1 forward to the farthest extent of the cutting edge, and parallel to the bucket floor. Because the operator must regularly check and, if required, update the cutting edge length, the exact location for J1 should be chosen in consultation with the machine operator.





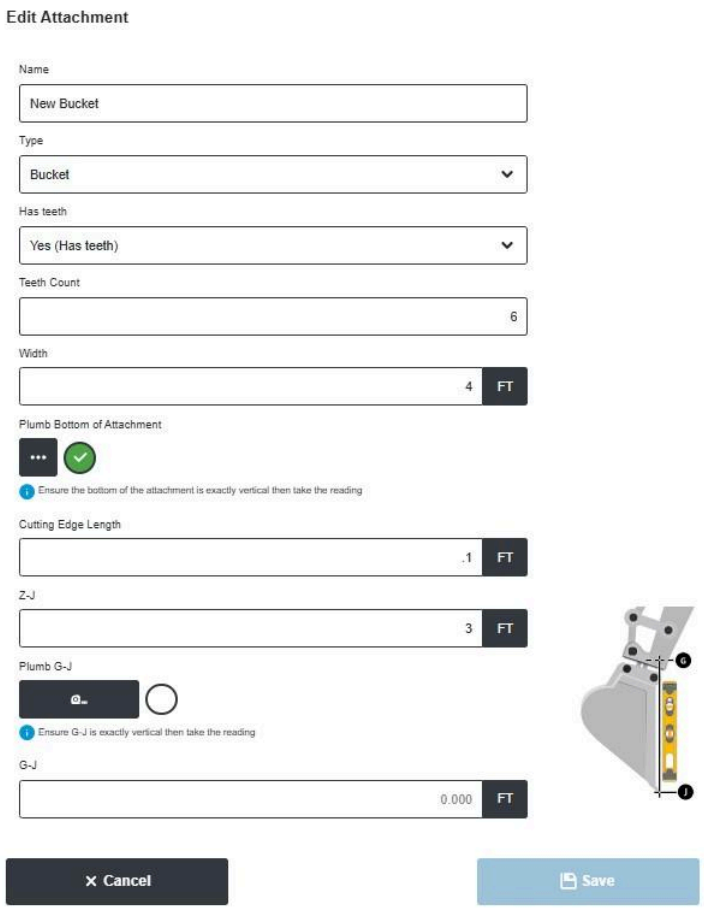
10. Measurement and enter the Z-J dimension. Use a tape measure to measure the farthest extent of the cutting edge(J) to the tangent point of the bucket floor(Z)

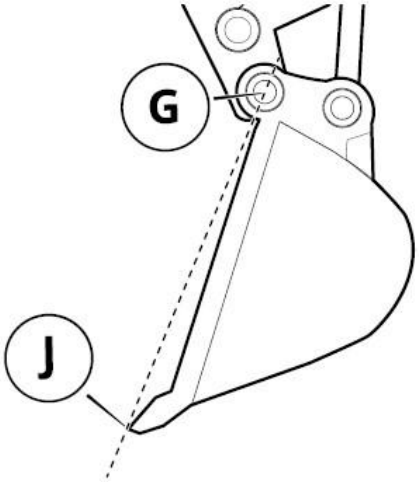




Note- Tangent is point T shown on the above diagram. The measurement point Z should be placed where the bucket transitions from flat to curved.

11. Plumb G-J - uncurl/curl the bucket so pins G-J are vertical as pictured in the graphic.

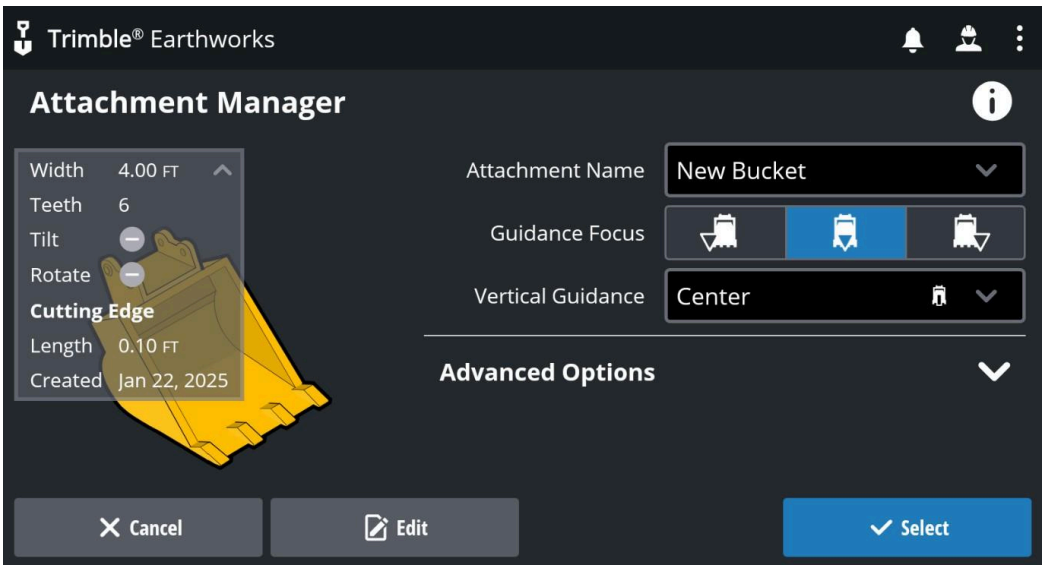




12. Enter the G-J measurement and select Save

The new attachment will now be listed among the attachments available in the web interface and operator interface.

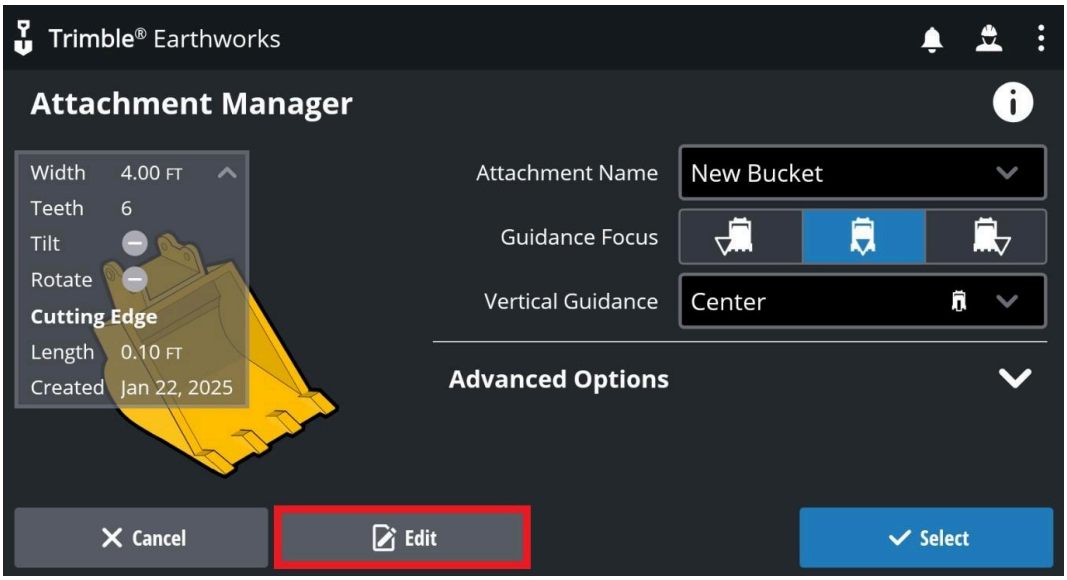
13. Return to the Operator App and enter the Attachment Manager screen via the Machine Setup Tile. Verify that the bucket is available to the operator.



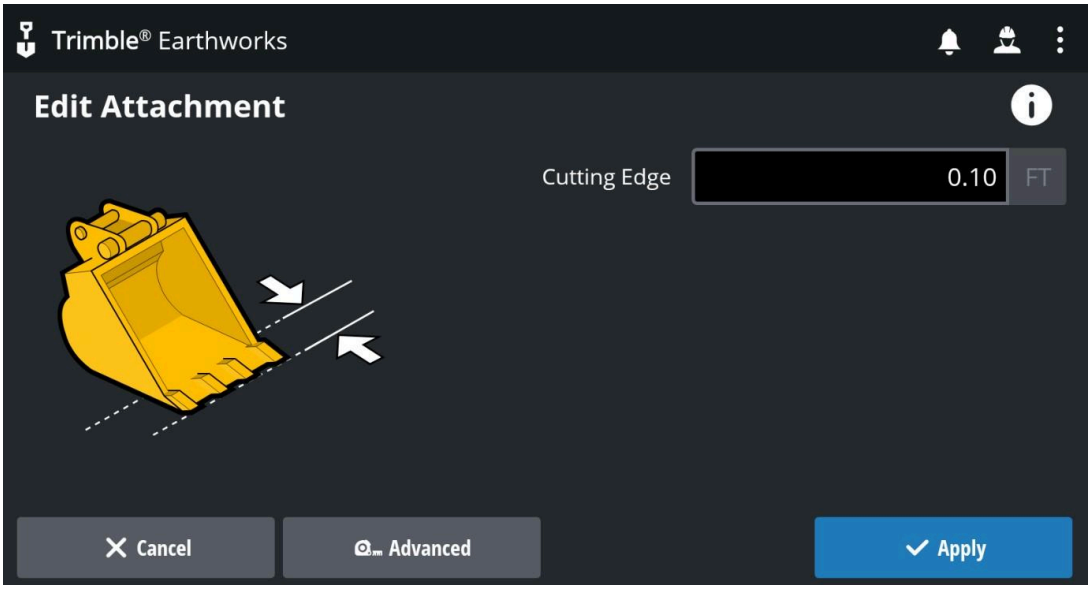
Editing from Earthworks Application

During the course of normal operation, the cutting edge of the bucket will wear. Wear of the teeth or cutting edge can introduce inaccuracy in calculated position if it is not periodically adjusted. This is accomplished via the Attachment Manager screen.

14. Enter the Operator App and navigate to the Attachment Manager via the Machine Setup Tile. Select the Edit button.



15. Physically measure the Cutting Edge Length of the bucket and enter the dimension.

Note- Because the operator must regularly check and, if required, update the cutting edge length, the exact location for J1 should be chosen in consultation with the machine operator. It is imperative that the cutting edge be measured using the same points used in the initial measure up to maintain accuracy of the attachment.

16. Click Apply.